From Visual Cues to Spoken Narration: Rethinking Audio Description

Akshita Gupta[1] **Aditya Arora**[1] **Federico Tombari**[2,3]
Marcus Rohrbach[1] **Anna Rohrbach**[1]

[1]TU Darmstadt & hessian.AI, Darmstadt, Germany
[2]Google Research [3]TU Munich
akshita.gupta@tu-darmstadt.de

Abstract

Audio Description (AD) provides spoken narration of visual events during dialogue gaps, making movies accessible to visually impaired audiences. The problem requires determining both *what* (which visual event) and *when* (position for inserting the AD) to narrate, to achieve the best user experience. Prior work has largely reduced the problem to video captioning of pre-segmented video clips, i.e., *what* is largely predefined and *when* is ignored entirely. We propose Cue2Narrate, a two-stage pipeline that jointly predicts *what* and *when* to narrate in longer untrimmed movie clips. A dual-head audio-visual localizer predicts two temporally distinct windows per AD utterance: a visual cue window and a spoken narration window. A LoRA-adapted VLM then generates concise ADs from the predicted visual evidence, trained with a Description Ranking Loss that ranks captions (negative samples) of the same frames lower than the GT AD. To benchmark this new problem statement, we introduce the LongLSMDC benchmark with up to 8-min movie clips (∼6.5 min on average). On LongLSMDC, Cue2Narrate outperforms video-only and audio-only localization baselines by 5–12 points in avg. mAP. Under both predicted- and GT-window evaluation, Cue2Narrate improves AD generation over the corresponding fine-tuned base VLM. These results establish the first benchmark for multi-segment AD generation on long-form clips. Data & Code: multimodal-ai-lab/Cue2Narrate

1 Introduction

Audio Description (AD) is spoken narration that makes movies accessible to blind and low-vision audiences by describing on-screen content not conveyed through dialogue (Benecke, 2007). A trained human describer watches the film, identifies the visual events that matter to the story (actions, expressions, scene changes, character interactions), and writes short descriptions to be voiced over the soundtrack (Packer et al., 2015; DCMP, 2020; Pavel et al., 2020). Since AD is mixed into the existing audio, each description must ideally be inserted into a dialogue "gap", avoiding overlap with speech (Pavel et al., 2020). The describer therefore makes two distinct decisions each time: *what* visual event to describe, and *when* the description should be voiced. The latter is typically done in the gap that occurs during or slightly before/after the event itself (Pavel et al., 2020; Branje and Fels, 2012; Fryer, 2016). However, professional AD production is slow and expensive, costing roughly $15–$75 per finished minute, limiting accessibility to a small fraction of released films (3Play Media, 2022, 2025); automating this process is essential to scale AD coverage. Most prior works on automatic AD generation significantly reduce the problem. First, they often evaluate in an oracle-window setting: the model is given the temporal interval for which a description should be produced (Rohrbach et al., 2017; Han et al., 2023b, 2024; Fang et al., 2025; Xie et al., 2025). Namely, AD boils down to video captioning over pre-defined short clips. Besides, prior works ignore the distinction between the *what* and *when* intervals. Existing AD datasets provide one type of temporal windows: CMD-AD and TV-AD (Han et al., 2024; Xie et al., 2024) align AD text to AudioVault[1] *spoken* slots, while LSMDC and MAD-Eval (Rohrbach et al., 2017; Han et al., 2023b) provide *visually*-aligned short clips with text. Prior works do not highlight this distinction. To sum up, most evaluate *what to say* given an oracle window without the challenge of long untrimmed video, and without predicting *when to actually speak*. The few works that touch on description timing inherit these limitations: AutoAD-II (Han et al., 2023a) binary-classifies

[1]AudioVault is a community-maintained non-profit repository of movie ADs. See https://audiovault.net.

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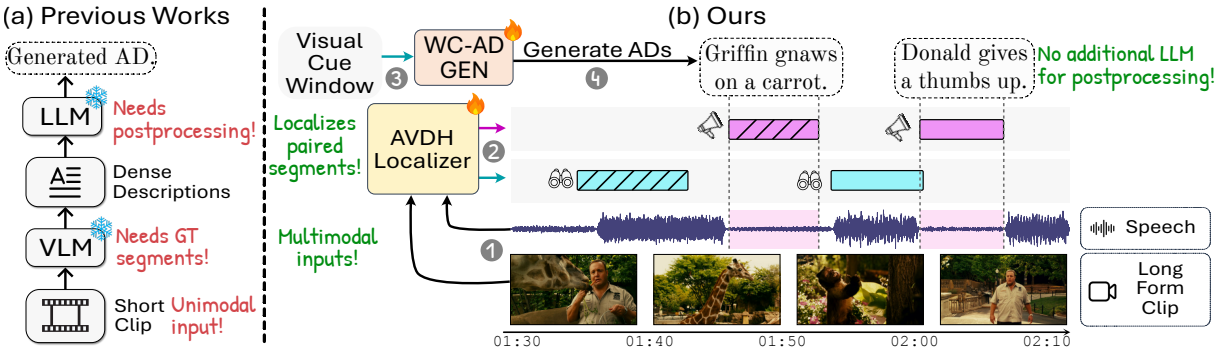


Figure 1: **Cue2Narrate overview.** **(a)** Prior methods caption GT clips with a VLM and then refine the dense descriptions with an LLM into AD-style text. **(b)** Given a long-form clip and the corresponding audio, our AVDH Localizer (Cue2Narrate) predicts paired temporal windows per AD: a visual cue window marking when the event is on screen, and a spoken narration window marking the dialogue gap in which the AD is inserted. Our WC-AD Gen (Cue2Narrate) generates a description from the localized visual window, to be inserted into the predicted spoken window. The figure shows an actual example from LongLSMDC, where these windows are routinely offset.

pre-supplied speech gaps, and CA3D (Lee et al., 2025) detects a single event per short clip. Thus, the full AD problem, with visual/spoken cue windows in untrimmed video with multi-segment prediction, has not been tackled so far.

We address these limitations twofold. First, we introduce **LongLSMDC**, a long-form AD benchmark with up to 8-min movie clips (∼6.5 min on average) with dense multi-segment annotations, built on top of LSMDC (Rohrbach et al., 2017, 2021). Unlike prior benchmarks, LongLSMDC provides both types of windows per AD utterance, matching how human describers produce AD: the visual cue window is paired with the spoken narration window in which it is voiced. Second, we propose **Cue2Narrate** (Fig. 1), a paired-window AD pipeline for untrimmed video: a dual-head audio-visual localizer (AVDH Localizer) predicts both windows (visual and spoken) per AD utterance, and WC-AD generator produces the description from the predicted localized visual cue window, to be inserted into the spoken narration window. On LongLSMDC, our localizer outperforms a video-only ActionFormer (Zhang et al., 2022) baseline by 11.8 points in avg. mAP, an audio-only ActionFormer variant by 5.0 points, and a dual-head ActionFormer baseline by 3.6 points. We also show transfer to CMD-AD and MAD-Eval, where we evaluate on the respective temporal windows. In terms of AD quality, Cue2Narrate improves CIDEr over its base VLM on both LongLSMDC and CMD-AD under both predicted- and GT-window evaluation. Generated ADs are matched to GT via tIoU on the predicted spoken-narration windows, and to our knowledge ours is the first protocol to jointly evaluate *what* and *when* in AD generation.

2 Related Work

Audio Description Generation.

Automated AD generation has been studied almost entirely as a text-generation problem given a temporal window. AutoAD (Han et al., 2023b) introduces VLMs for AD generation from a known interval, leveraging text-only AD corpora and visual captioning datasets for pretraining; AutoAD-II (Han et al., 2023a) extends this with character recognition and binary classification of ASR-extracted speech gaps to decide where AD should be inserted, pretraining on AudioVault-AD text and WebVid video-text (Bain et al., 2021); AutoAD-III (Han et al., 2024) releases CMD-AD with AudioVault-aligned spoken slots and introduces HowToAD-style long-video pretraining. Subsequent work focuses on richer generation conditioned on the assumed window, often with external character banks or movie scripts: MM-Narrator (Zhang et al., 2024), NarrAD (Park et al., 2025), UniAD (Wang et al., 2025), AutoAD-Zero (Xie et al., 2024), Shot-by-Shot (Xie et al., 2025), and DistinctAD (Fang et al., 2025). The corresponding benchmarks preserve only one part of the "paired" AD annotation: CMD-AD and TV-AD (Han et al., 2024; Xie et al., 2024) contain *spoken* slots without a visual cue window, while LSMDC and MAD-Eval (Rohrbach et al., 2017; Han et al., 2023b) contain *visual* slots without spoken narration window. Parallel work (Kala et al.,

Table 1: **Audio description datasets** compared across the dimensions needed to predict both AD windows on long-form clips. Prior datasets are either short-clip (LSMDC, MAD-Eval) or contain spoken-only segments on longer clips (CMD-AD, TV-AD; frames available by source). LongLSMDC is the first to support multi-segment prediction of the visual and spoken windows on long-form clips.

Dataset	Source	Domain	#Annotations	Clip length	ADs/clip	Visual	Spoken
LSMDC (Rohrbach et al., 2017)	LSMDC	Movies	108.5K	∼5 sec	1	✓	✗
MAD-Eval (Han et al., 2023b)	MAD (LSMDC subset)	Movies	7K	∼5 sec	1	✓	✗
CMD-AD (Han et al., 2024)	CMD + AudioVault	Movies	101K	∼2 min	∼10	✗	✓
TV-AD (Xie et al., 2024)†	TV + AudioVault	TV	34K	unknown	–	✗	✓
LongLSMDC (ours)	LSMDC full movies	Movies	83.8K	∼6.5 min	∼40	✓	✓

† TV-AD releases only pre-extracted frames, so clip-length statistics cannot be computed.

2025) also operate on few-minute video segments rather than trimmed clips, but focus on QA-based evaluation of *generated* ADs rather than predicting AD timing. None support joint prediction , and the closest timing-related work, CA3D (Lee et al., 2025), predicts a single window per clip rather than a paired structure. We instead predict the pairs in our Stage 1 on long-form clips by using video and audio modalities, supervised by LongLSMDC annotations. Where prior methods generate from an assumed or oversampled visual context, we learn to predict the visual cue window and pair the generated description with the predicted spoken narration window, mirroring how human describers produce AD.

Temporal Localization and Dense Captioning. Temporal action localization (TAL) identifies multi-segment temporal boundaries in untrimmed video with transformer-based architectures (Zhang et al., 2022; Shi et al., 2023) and cross-modal cues (Tian et al., 2018; Senocak et al., 2018, 2023). Dense video captioning extends this to producing descriptions per segment, either as separate (Krishna et al., 2017) or end-to-end (Iashin and Rahtu, 2020; Li et al., 2018; Yang et al., 2023; Islam and Bertasius, 2022) pipelines. Both lines predict visual-only segments anchored to scene changes or action boundaries, with no requirement on the audio track. AD differs structurally: the output is a paired (visual, spoken) annotation, where the spoken window is a dialogue gap that may sit before, during, or after the event being described (Pavel et al., 2020). The visual frame at an AD's spoken window therefore need not match the description's text, placing AD outside the scope of visual-only localization and captioning. Our audio-visual dual-head localizer predicts both windows of the pair, with each head supervised against its own ground-truth window.

3 LongLSMDC Dataset

Our LongLSMDC benchmark is constructed from the full-length movies in LSMDC/LSMDC v2 (Rohrbach et al., 2017, 2021). The commonly used annotations only provide timestamps for visual cue windows. Importantly, we were able to get access to the corresponding spoken narration windows.[2] We segment each movie into non-overlapping clips of up to 8 min (∼6.5 min), assign each AD annotation to the clip containing its temporal midpoint, and convert movie-level timestamps to clip-relative coordinates. The resulting dataset contains 1,995 training clips and 170 validation clips, with an average of ∼40 AD annotations per clip (Table 1).

Training annotations span three supervision categories: 37,302 spoken-visual window (y^{vn} and y^{sp}), 21,295 visual-only (y^{vn}), and 20,197 spoken-only (y^{sp}) (78.8k training annotations; 83.8k including validation, Table 1).[3] Within the paired set, 32,715 (∼88%) exhibit a temporal offset between the two windows, confirming that the visual and spoken windows are routinely distinct. The validation set contains 5,006 paired annotations, supporting evaluation against both windows. We additionally create a per-clip character bank, and action-event annotations from each AD sentence via a closed-source model (Comanici et al., 2025) to support the Action Score metric (Xie et al., 2025). More details are provided in Appendix A.

[2] Obtained from the original authors.

[3] This is due to various post-processing steps that occured between extraction of the spoken windows and manual alignment to the visual windows carried out in LSMDC.

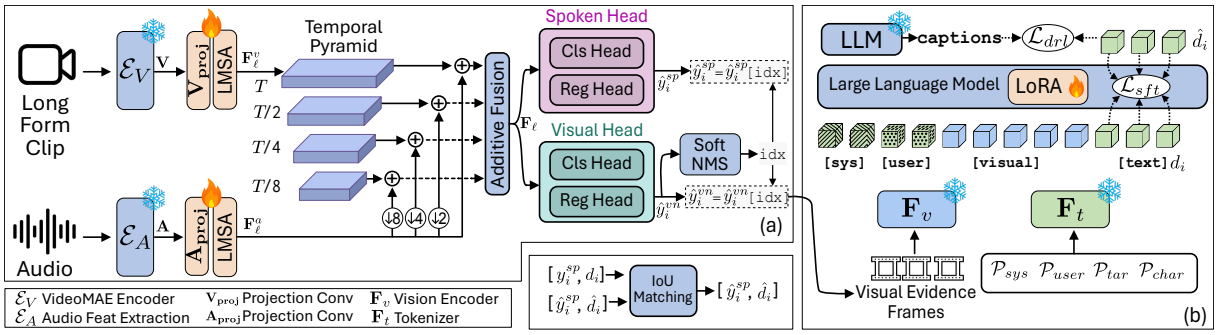


Figure 2: **Cue2Narrate architecture. (a) Localization (Stage 1).** Frozen video ($\mathcal{E}_V$) and audio ($\mathcal{E}_A$) encoders extract features that are projected, refined with self-attention (LMSA), and encoded into a 4-level temporal pyramid ($T, T/2, T/4, T/8$). The streams are combined by additive fusion. Two independent heads the Spoken Head and Visual Head, each with classification and regression branches predict the spoken narration window and visual cue window, respectively. Soft-NMS yields final segments; predicted spoken windows ($\hat{y}_i^{sp}[\texttt{idx}]$) drive evaluation matching, while predicted visual windows ($\hat{y}_i^{vn}[\texttt{idx}]$) condition Stage 2. **(b) Generation (Stage 2).** For each predicted visual cue window, visual evidence frames are encoded into $\mathbf{F}_v$. A structured prompt $\mathcal{P} = [\mathcal{P}_{sys}, \mathcal{P}_{user}, \mathcal{P}_{target}, \mathcal{P}_{char}]$ fed to tokenizer $\mathbf{F}_t$. A frozen LLM with LoRA-adapted projection is trained with $\mathcal{L}_{sft}$ on ground-truth ADs and $\mathcal{L}_{drl}$ against a caption sampled from a frozen LLM on the same frames, generating the AD sentence $\hat{d}_i$. **(c) Evaluation.** Under predicted windows, generated ADs must be matched to ground-truth annotations before scoring; we match on spoken narration window tIoU because that is the side AD references are aligned to in human annotation practice.

4 Cue2Narrate

4.1 Problem Formulation

Let $X = (\mathcal{V}, \mathcal{A})$ denote a long-form clip with video $\mathcal{V}$ and audio $\mathcal{A}$. The target output is a set of AD annotations

$$\mathcal{Y} = \{(d_i,\, y_i^{vn},\, y_i^{sp})\}_{i=1}^{N}, \tag{1}$$

where d_i is the AD sentence, $y_i^{vn} = [vn_s^i,\, vn_e^i]$ is the visual cue window interval in which the described event is visible on screen, and $y_i^{sp} = [sp_s^i,\, sp_e^i]$ is the spoken narration window interval in which d_i is delivered during a dialogue gap. At test time, both the temporal windows and the sentence content are unknown, as well as the number of AD annotations N. Our method factorizes this problem into two stages. Stage 1 predicts paired temporal windows $(\hat{y}_i^{vn}, \hat{y}_i^{sp})$ from the raw clip X. Stage 2 generates a sentence $\hat{d}_i$ from the visual evidence inside the predicted visual cue window, $\hat{y}_i^{vn}$. The final delivered AD is $(\hat{d}_i, \hat{y}_i^{sp})$: the visual cue window determines *what* is described, and the spoken narration window determines *when* that description is inserted.

4.2 Stage 1: Audio-Visual Dual-Head (AVDH) Localizer

Stage 1 localizes AD slots in a long-form clip by predicting, for each instance, a visual cue window y_i^{vn} and a spoken narration window y_i^{sp}. In LongLSMDC, 88% of paired annotations exhibit a temporal offset between these two windows; the visible event and the spoken narration differ in onset, duration, or both. We predict them with two independent heads over a shared audio-visual representation, each supervised against its own GT windows. We extract visual features $\mathbf{V} \in \mathbb{R}^{T\times 768}$ with a VideoMAE encoder (Tong et al., 2022) and audio features $\mathbf{A} \in \mathbb{R}^{T\times 130}$, a 130-dimensional log-mel spectrogram, resampled to the visual timeline. Fig. 2 (a) The audio stream directly exposes speech activity and silence patterns, critical cues for predicting when a narration slot is available. The visual features are projected and refined with temporal self-attention, then encoded into a multi-scale temporal pyramid $\{\mathbf{F}_\ell^v\}_{\ell=1}^{L}$ via a ConvTransformer backbone, covering AD events of different durations. The audio features are projected to d_h=256, refined with lightweight temporal self-attention, and average-pooled to each pyramid level, yielding $\{\mathbf{F}_\ell^a\}_{\ell=1}^{L}$. We fuse the two modalities by pyramid-level addition:

$$\mathbf{F}_\ell = \mathbf{F}_\ell^v + \mathbf{F}_\ell^a. \tag{2}$$

This design is motivated by the complementary strengths of the two modalities (Figs. 3, 4): audio dominates spoken narration window localization where silence patterns are the primary cue, while video dominates visual cue window localization where scene changes and character actions matter.

Both heads operate on the fused pyramid $\{\mathbf{F}_\ell\}_{\ell=1}^{L}$ but predict independently. Each head contains a focal classification branch (Lin et al., 2017) for foreground/background scoring and a regression branch that predicts non-negative left/right offsets (δ_l, δ_r) from each temporal anchor t. For pyramid level ℓ with stride s_ℓ, the predicted interval is

$$[t - \delta_l \cdot s_\ell,\ t + \delta_r \cdot s_\ell]. \tag{3}$$

The visual cue window head predicts y_i^{vn}; the spoken narration window head predicts y_i^{sp}. This allows the model to represent windows with different durations and arbitrary temporal offsets. For spoken classification, we add a learned temporal-position prior over normalized clip time, giving the head a coarse prior over where narration slots tend to occur in a clip. Further, Soft-NMS (Bodla et al., 2017) suppresses duplicate predictions by down-weighting overlapping candidates by their tIoU rather than removing them.

Training objective. Both heads are trained jointly:

$$\mathcal{L}_{\text{stage1}} = \mathcal{L}_{\text{cls}}^{vn} + \mathcal{L}_{\text{reg}}^{vn} + \lambda_{sp}\big(\mathcal{L}_{\text{cls}}^{sp} + \mathcal{L}_{\text{reg}}^{sp}\big), \tag{4}$$

with λ_{sp}=1. Both classification terms use sigmoid focal loss (Lin et al., 2017); both regression terms use a 1D center-based DIoU loss. LongLSMDC annotations supervise the heads asymmetrically: visual cue window-only annotations train only the visual head, spoken narration window-only annotations train only the spoken head, and spoken-visual window annotations train both. The paired subset is the key supervision that prevents the two heads from collapsing to a single shared window.

4.3 Stage 2: Window-Conditioned AD Generator (WC-AD GEN)

Stage 2 maps a localized visual cue window to a single-sentence audio description, Fig. 2 (b). Beyond describing the correct event, the generator must produce narration in the concise AD style rather than captioning the whole clip in great detail. Standard cross-entropy fine-tuning teaches the model *what* to say but not *how*: AD training data is too sparse to overwrite the pretraining prior towards long, descriptive captions. We address this with a Description Ranking Loss (DRL) that suppresses the caption prior during training. We build Stage 2 on Qwen models (Bai et al., 2025) as the backbone, adapting it with LoRA (r=16, α=32, dropout 0.05) on the language model and multimodal projection while keeping the vision encoder frozen. Frames are sampled uniformly from the visual cue window (F=32 at training, F=64 at inference) and encoded by the frozen vision encoder, then aligned via a trainable projection layer. For each annotation i, the model receives a prompt $\mathcal{P}_i$ containing: (i) a task instruction for one-sentence AD generation, (ii) visual tokens from y_i^{vn}, (iii) optional character-bank identities, and (iv) target-window context. The spoken narration window is not used as visual evidence; it determines position of the generated sentence.

Loss functions. Let $y_i^+ = d_i$ denote the ground-truth AD and $\mathbf{V}_i$ the sampled frames from y_i^{vn}. The supervised fine-tuning loss is

$$\mathcal{L}_{\text{sft}} = -\sum_t \log P(y_{i,t}^+ \mid y_{i,<t}^+, \mathbf{V}_i, \mathcal{P}_i). \tag{5}$$

We additionally compute a ranking loss to suppress the caption prior. For each training sample, a negative y_i^- is precomputed by prompting a frozen LLM on $\mathbf{V}_i$ with a simple prompt (no AD framing) and cached offline. We compute mean per-token log-likelihoods for the ground-truth and the negative under the same model and prompt:

$$\begin{aligned} s_i^+ &= \tfrac{1}{|y_i^+|}\sum_t \log P(y_{i,t}^+ \mid y_{i,<t}^+, \mathbf{V}_i, \mathcal{P}_i), \\ s_i^- &= \tfrac{1}{|y_i^-|}\sum_t \log P(y_{i,t}^- \mid y_{i,<t}^-, \mathbf{V}_i, \mathcal{P}_i), \end{aligned} \tag{6}$$

where prompt tokens and padding positions are excluded by the label mask. DRL encourages higher normalized likelihood for the concise AD than for the "vanilla" caption:

$$\mathcal{L}_{\text{drl}} = -\log \sigma\left(\tfrac{s_i^+ - s_i^-}{\tau}\right). \tag{7}$$

Since the negative is generated by the frozen VLM, DRL directly targets the residual pretraining prior that cross-entropy alone cannot suppress. Thus, the full Stage 2 objective is:

$$\mathcal{L}_{\text{stage2}} = \mathcal{L}_{\text{sft}} + \lambda\,\mathcal{L}_{\text{drl}}, \qquad \lambda{=}0.2,\ \tau{=}0.5. \tag{8}$$

During training, Stage 2 uses ground-truth visual cue window s; at inference, predictions from Stage 1 are used instead.

4.4 Joint Inference

The two stages couple naturally at test time: Stage 1 predicts *where* to look and narrate, Stage 2 generates exactly *what* to say, the same decisions

Table 2: **Stage 1 localization and transfer.** Average mAP over tIoU 0.2–0.6. All methods are trained on LongLSMDC. **Visual** and **Spoken** report each head's mAP against its own GT window; **Avg.** is the mean. ActionFormer rows are single-head configurations; *Baseline* is a two-head ActionFormer receiving the same audio-visual input with naive modality combination. Transfer to CMD-AD reports spoken narration window mAP and to MAD-Eval visual cue window mAP. ZS = zero-shot; FT = fine-tuned on the target dataset. "–" indicates not evaluated or not predicted.

Method	Modality		LongLSMDC			CMD-AD		MAD-Eval
	Video	Audio	Visual	Spoken	Avg.	FT	ZS	ZS
Spectrogram	✗	✓	1.4	1.3	1.4	1.7	1.7	23.1
Gemini 2.5 Pro	✓	✓	7.1	4.4	5.8	–	–	–
ActionFormer	✓	✗	32.0	28.5	30.3	–	–	63.7
Audio-ActionFormer	✗	✓	35.2	39.1	37.1	39.2	25.0	–
Baseline	✓	✓	39.0	38.1	38.5	–	–	–
Cue2Narrate	✓	✓	**42.2**	**42.0**	**42.1**	**41.0**	**26.5**	**70.0**

a human describer makes. At test time, Stage 1 predicts paired windows $\{(\hat{y}_i^{vn}, \hat{y}_i^{sp})\}_{i=1}^{\hat{N}}$ from the raw clip. For each pair, Stage 2 generates $\hat{d}_i$ from $\hat{y}_i^{vn}$ alone. No previously generated AD is supplied as context ($k = 0$); the phrase refers only to standard autoregressive decoding within a single sentence. The final output is:

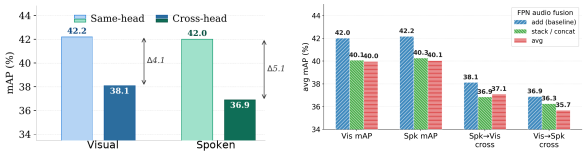
$$\{(\hat{d}_i,\ \hat{y}_i^{sp})\}_{i=1}^{\hat{N}}, \tag{9}$$

with *no ground-truth boundaries used at any point*. The total cost is one Stage 1 forward pass and one VLM decoding per AD slot.

5 Experiments

Datasets. For Stage 1, we train on LongLSMDC and evaluate on LongLSMDC, CMD-AD (Han et al., 2024), and MAD-Eval (Han et al., 2023b). For Stage 2, we only use the datasets with spoken narration window's annotations. More dataset and metric details in Appendix A.

Joint evaluation. When the AD Generator is conditioned on Stage 1 predictions, Stage 2 generates AD text from predicted visual cue window, while temporal matching for metrics is performed on the paired predicted spoken narration window. Concretely, the predicted visual cue window drives AD generation only, while the predicted spoken narration window drives temporal matching to GT spoken annotations. A prediction is matchable only against GT spoken segments from the same video with tIoU > 0.3; among multiple matches to the same GT segment, greedy



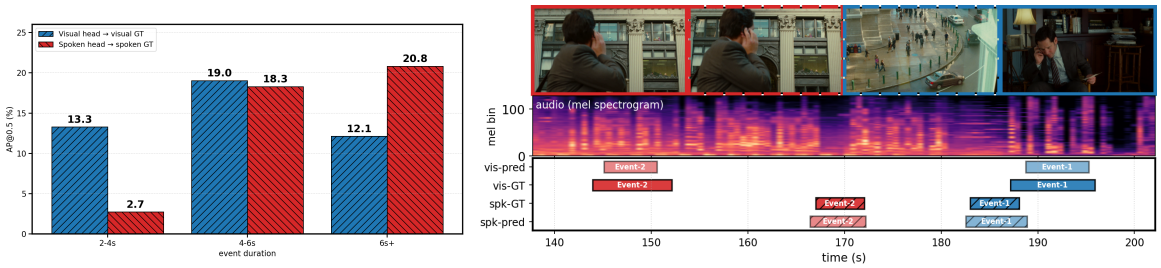
(a) (b)

Figure 3: **Stage 1 Dual-Head Analysis.** (a) Each head evaluated against its own GT (**Same-head**) and against the other head's GT (**Cross-head**); the ∼4–5 point drop confirms the visual and spoken heads learn distinct temporal targets rather than a shared window. (b) Audio-fusion strategies (add, stack, avg) compared on visual mAP, spoken mAP, and the two cross-head settings. Element-wise addition consistently wins, justifying our fusion choice.

one-to-one matching retains the highest tIoU. Text metrics (CIDEr (Vedantam et al., 2015), Action Score (Xie et al., 2025), Recall (Han et al., 2023b), LLM-Eval (Han et al., 2024; Zhang et al., 2024)) are computed on matched pairs only, reducing false positives from unmatched localization errors and keeping evaluation aligned with the spoken-side annotation protocol.

5.1 Localization Results

Table 2 reports localization on LongLSMDC with zero-shot transfer to CMD-AD and MAD-Eval; we also report the fine-tuning results for the former; no additional pretraining data is used. Cue2Narrate achieves 42.2 visual and 42.0 spoken mAP on LongLSMDC, outperforming video-ActionFormer by 11.8 points and the audio-only variant by 5.0 points. The gain over ActionFormer confirms that audio is a critical modality: silence patterns directly signal narration slot availability, information unavailable to a video-only model. The residual gap between audio-only and the full model reflects the complementary role of video, which identifies which silences correspond to narratively salient events. On CMD-AD, Cue2Narrate reaches 41.0 spoken mAP (FT), above *Audio*-ActionFormer (39.2). On MAD-Eval, it achieves 70.0 visual mAP, above ActionFormer (63.7). CA3D (Lee et al., 2025), the only prior AD localizer, reports 65.3 F1 on MAD-Eval; our model reaches 90.0 F1 on the same split (details in Appendix B.1), confirming the gain is not a metric artifact. Furthermore, to have fair multi-modal baseline comparison, we extend ActionFormer with a dual-head architecture and naive audio-visual fusion which



(a) (b)

Figure 4: **Modality Contribution.** (a) Per-instance AP@0.5 on paired annotations bucketed by event duration, for the visual head against the visual GT window (blue) and the spoken head against the spoken GT window (red). (b) Qualitative example with mel-spectrogram and frame strip with vis-pred/vis-GT and spk-pred/spk-GT tracks, illustrating the temporal offset between paired visual and spoken windows that Cue2Narrate recovers.

gives 38.5 average mAP. As a closed-source ablation, we probe Gemini 2.5 Pro (Comanici et al., 2025) for localization via prompted timestamp prediction; despite its strong video understanding, it fails to produce reliable temporal boundaries, highlighting that precise AD localization requires dedicated training (per-tIoU breakdowns in Appendix B). All Cue2Narrate results are means over three seeds, with standard deviation at most 0.2 mAP. The complete three-seed localization results are reported in Appendix Table 6.

Dual-head design and fusion strategy. We ablate the two key design choices: The dual-head structure and how the audio stream is fused into the visual FPN. Fig. 3a shows the heads are not redundant: each is strong on its own target but degrades by 4–5 points when evaluated against the other head's GT. The drop reflects the fact that paired visual and spoken windows in LongLSMDC are routinely offset in time (Sec. 3); a single shared head would be forced to predict one window or an average of the two, losing the offset that defines AD timing. The dual-head design preserves it. Fig. 3b compares three strategies for combining the audio stream with the visual FPN: addition (ours), stack/concat, and avg. Addition wins consistently across all four settings, with concat and avg trailing by 1.5–2 mAP. Concat forces the model to learn a new joint representation at each pyramid level, diluting the modality-specific signal; avg dampens both streams symmetrically. Addition keeps each modality's contribution intact and lets each head exploit them independently, audio for silence cues, video for visual salience, which is exactly the complementarity the dual-head design was set up to capture.

Per-instance and qualitative evidence: each modality dominates its regime. Audio reliably finds the dialogue gaps where narration can be inserted but cannot say which gaps are worth narrating; video supplies the complementary signal — shot changes, character actions, and on-screen events that distinguish a narratable moment from an arbitrary pause. Fig. 4a shows this split at the instance level on paired annotations bucketed by event duration. The spoken head (red) is weak at 2–4 s (AP@0.5 = 2.7) where dialogue gaps are too short to disambiguate but dominates at 6 s+ (20.8) where the gap structure is unambiguous: Its strength scales with the extent to which the gap is observable. The visual head (blue) is comparatively flat across durations (13.3 vs. 12.1). Since visual evidence accumulates over scene context rather than gap length, a short and a long event can carry a comparable visual signal. The two modalities cover non-overlapping regimes. Fig. 4b makes this concrete on a paired clip with two events: the mel-spectrogram shows a dialogue dip in each, which the spoken head (`spk-pred`) aligns with, while the visual head (`vis-pred`) locks onto the on-screen action visible in the frame strip. The visual cue window and spoken narration window for the same event are temporally offset; the action and its narration do not coincide, and Cue2Narrate recovers both sides of the offset.

End-to-end coverage. Table 5 evaluates predicted spoken windows using greedy one-to-one matching. Precision is the fraction of predictions matched to a GT spoken narration window, recall the fraction of GT spoken windows matched, and F1 (mean). Accordingly, unmatched predictions equal $1 - P$ and missed narration slots $1 - R$.

5.2 Generation Results

We further extend to the *what* component of AD, generating the sentence that describes the event.

Table 3: **Stage 2 generation on LongLSMDC and CMD-AD.** Results use ground-truth spoken narration window segments; LongLSMDC is evaluated on **paired data** with spoken narration window, and CMD-AD uses CharBank when available. Code is unavailable for AutoAD-III and DistinctAD, so only their reported CMD-AD numbers are included. TF = training-free; C = CIDEr; AS = Action Score; R@1/5 = Recall@1/5. †UniAD is run on LongLSMDC from the released MAD-pretrained checkpoint.

Method	TF	LongLSMDC			CMD-AD (Zero-shot)		
		C	AS	R@1/5	C	AS	R@1/5
AutoAD-III	✗	–	–	–	25.0	31.5	31.2
DistinctAD	✗	–	–	–	22.7	–	33.0
UniAD†	✗	10.1	–	–	21.8	–	–
AutoAD-Zero	✓	14.2	–	31.3	17.7	25.5	26.9
Shot-by-Shot	✓	16.9	25.3	28.4	**26.3**	28.4	33.0
Qwen2.5-VL-7B	✗	30.1	41.7	43.8	20.5	35.4	34.0
+ **Cue2Narrate**	✗	32.5	40.9	43.3	22.3	35.0	33.2
Qwen3-VL-8B	✗	36.0	42.0	46.8	23.8	35.1	34.1
+ **Cue2Narrate**	✗	**37.3**	**42.6**	**47.0**	25.1	**36.0**	**34.4**

Table 4: **Stage 1+2 generation on LongLSMDC and CMD-AD.** Results are computed under Stage 1 predicted segments using tIoU@0.3 matching. Results on paired data. TF = training-free; C=CIDEr; AS=Action Score; R@1/5=Recall@1/5.

Method	TF	LongLSMDC			CMD-AD (Zero-Shot)		
		C	AS	R@1/5	C	AS	R@1/5
AutoAD-Zero	✓	5.1	20.7	23.4	10.1	**30.9**	25.3
Qwen2.5-VL-7B	✗	21.7	38.8	37.3	7.5	23.4	24.8
+ **Cue2Narrate**	✗	22.2	38.3	37.0	9.2	25.1	24.9
Qwen3-VL-8B	✗	23.8	39.4	39.3	8.8	24.8	25.2
+ **Cue2Narrate**	✗	**25.2**	**40.2**	**40.1**	**11.5**	27.8	**26.0**

Baseline convention. In Tables 4 and 3, the Qwen2.5-VL-7B and Qwen3-VL-8B rows report numbers *after SFT on LongLSMDC AD annotations* (i.e, the same LoRA adaptation, prompt, and training data as Cue2Narrate, but with $\mathcal{L}_{sft}$ only). The `+Cue2Narrate` rows add $\mathcal{L}_{drl}$ on top of this SFT baseline. In all Stage 2 experiments, each AD is generated independently from its corresponding visual cue window; neither previously generated nor ground-truth neighbouring descriptions are provided as context.

Generation under predicted and GT spoken windows. Tables 4 and 3 report generation quality in two settings. We evaluate on paired spoken annotations. **S1→S2** (Table 4) Cue2Narrate generates from Cue2Narrate predicted visual windows and is scored against GT ADs matched through the predicted spoken window (Sec. 5). **GT→S2** (Table 3) replaces predicted windows with GT spoken segments, removing Cue2Narrate localization error from the evaluation and measuring generator quality in isolation. In both settings, Cue2Narrate improves over the base model on both backbones and both datasets: on LongLSMDC, +1.3 CIDEr for Qwen3-VL (36.0 → 37.3) under GT windows and +1.4 CIDEr (23.8 → 25.2) under predicted windows. For Qwen3-VL, the gain also holds for Action Score and R@1/5 and exceeds the seed-level std. by ∼4× (Table 7); for Qwen2.5-VL, it is limited to CIDEr. On CMD-AD, the gain transfers zero-shot (8.8 → 11.5 predicted; 23.8 → 25.1 GT). Separate three-seed generation stability results are provided in Appendix Tab. 7. For Qwen3-VL on LongLSMDC, CIDEr is 37.3 under GT windows and 25.2 under predicted windows. This difference should not be interpreted as a pure localization penalty because the predicted-window evaluation includes only successfully matched predictions, whereas the GT-window evaluation uses the complete paired GT set. On CMD-AD, the generator recovers when oracle input segments are provided (25.1 CIDEr) but drops sharply under predicted windows (11.5 CIDEr). Stage 1 still predicts a visual window for CMD-AD, but CMD-AD provides no GT visual annotations, so the generator was never exposed to CMD-AD's distribution of visual windows; the predicted visual windows come from a localizer trained on LongLSMDC, while CMD-AD clips are defined by narration-slot timing rather than the visual events. Thus CMD-AD serves as a transfer benchmark. Its released MAD-pretrained check-

Table 5: **End-to-end spoken-window localization coverage on LongLSMDC validation.** We use greedy one-to-one matching at tIoU thresholds 0.3 and 0.5.

tIoU	Precision	Recall	F1	Unmatched pred.	Missed slots
0.3	27.5	94.7	42.7	72.5%	5.3%
0.5	21.2	73.0	32.9	78.8%	27.0%

Method	C	LLM
Gemini-2.5 Pro	17.3	2.80
Qwen2.5 + Cue2Narrate	32.5	2.43
Qwen3 + Cue2Narrate	**37.3**	2.68

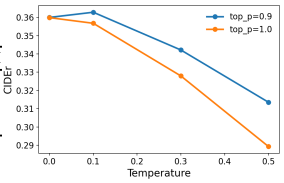


Figure 5: **Stage 2: LongLSMDC spoken windows. Left:** comparison against Gemini-2.5 Pro under GT→S2. **Right:** decoding-strategy ablation for Cue2Narrate; greedy decoding is on par with the best setting tested. All Stage 2 results use greedy decoding.



GT 1: Gordy pulls the hammer.

Pred 1: Ace stands and watches Chris walk away in the woods.

GT 2: Ace swallows and glances at Charlie.

Pred 3: Ace lowers his gaze and shifts his weight.

Figure 6: **Qualitative example on CMD-AD.** Top: sampled frames. Bottom: Pred segments (blue) vs. GT (black). GT/Pred AD sentences are listed below.

point under GT spoken windows gives 10.1 CIDEr on LongLSMDC, against 34.3 for Cue2Narrate (Qwen3-VL) in the matching no-CharBank setting; the 37.3 reported in Table 3 additionally uses CharBank. UniAD clearly outperforms the zero-shot Qwen3-VL backbone (5.3 CIDEr), confirming that the adaptation is faithful and that its contextual design transfers without target-domain training; the remaining gap reflects the distribution shift between UniAD's training data (short 2–3 min clips with sparse spoken-only slots) and LongLSMDC's dense, long-form clips with paired windows.

Scope of the matched-pair evaluation. Text metrics in Table 4 (S1→S2) are computed only on predictions that match a GT spoken window at tIoU > 0.3, since our proposed setting requires a visual–spoken pair to be present before an AD can be assessed against a reference. This mirrors how AD is produced in practice: an unmatched prediction has no reference sentence to score against. As a consequence, Table 4 reports quality *conditional on successful localization* and does not penalize unmatched predictions or missed GT segments; localization coverage is reported separately in Table 5. Table 3 (GT→S2) evaluates on the complete paired GT set and is directly comparable to prior SOTA numbers on CMD-AD. Because the two settings use different reference pools, their CIDEr values should not be interpreted as a direct estimate of the localization penalty.

Qualitative analysis on dense, long-form clips. Fig. 6 shows a CMD-AD clip with two AD events: Cue2Narrate recovers both spoken windows and generates contextually relevant descriptions, though phrasing diverges from the reference (e.g., *"Ace stands and watches Chris walk away"* vs. GT *"Gordy pulls the hammer"*) a common pattern when multiple plausible visual subjects share the frame. Dense, multi-event examples from LongLSMDC are in Appendix G. This substitution pattern is one of four failure modes we categorise in Appendix G: (i) wrong-subject attribution when multiple plausible visual subjects share the frame, (ii) correct event but an underspecified verb, (iii) boundary drift producing a description of adjacent content, and (iv) hallucinated detail. Representative examples are given there.

Comparison against closed-source models and decoding ablation. Fig. 5 probes two questions about Cue2Narrate. **Left:** how does Cue2Narrate compare to a strong closed-source VLM? Gemini-2.5 Pro (Comanici et al., 2025), evaluated in the **GT→S2** setting, trails both Cue2Narrate backbones on CIDEr by a wide margin (17.3 vs. 32.5 / 37.3) while scoring highest on LLM-Eval (2.80 vs. 2.43 / 2.68). The two metrics measure different properties: LLM-Eval rewards coherent, fluent narration, whereas CIDEr rewards n-gram overlap with the concise reference-AD style. Gemini produces fluent free-form descriptions that a judge model rates well, but does not conform to the compact AD register that Cue2Narrate's Description Ranking Loss explicitly trains for, which the CIDEr gap reflects. Tables 4 and 3 isolate the DRL contribution: the `+Cue2Narrate` rows, which add $\mathcal{L}_{drl}$ on top of the SFT baseline described above, consistently improve CIDEr, showing that DRL provides gains beyond standard fine-tuning. **Right:** how sensitive is Cue2Narrate to decoding choices? Greedy decoding is on par with the best setting tested, and CIDEr falls steadily as temperature increases, indicating Cue2Narrate's output distribution is sharp enough that sampling rarely finds better phrasings.

6 Conclusion

We re-framed automatic Audio Description as the joint prediction of *what* to narrate and *when* to narrate it, and introduced LongLSMDC to benchmark this problem on long-form movie clips. Cue2Narrate pairs a dual-head audio-visual localizer with a window-conditioned generator. The dual head separately predicts where the event is visible and where its narration is voiced, recovering the offset between the two that defines AD timing. We hope LongLSMDC will facilitate new exciting research in automatic AD generation.

7 Limitations

This work is a first step toward automatic audio description on long-form, untrimmed video, and opens up several directions. (i) Two-stage training. Our pipeline keeps localization and generation separate; jointly training the two stages would let generation feedback shape window prediction and is a natural next step. (ii) Cross-description context. Stage 2 conditions on the visual cue window and generates each AD independently. Extending the generator to reason over the spoken narration window and adjacent ADs (surrounding dialogue, previously voiced descriptions) could further improve coherence across consecutive descriptions. (iii) User study. A study with blind and low-vision users would complement our 20-evaluator pairwise study of coverage, timing, and narrative quality, and we leave it to future work as it is beyond the scope of this paper. (iv) Character banks. Character identity is assigned automatically; Appendix A quantifies the pipeline's agreement rate, and human verification of the banks is a natural extension.

8 Ethical Considerations

Cue2Narrate is intended as an assistive tool to expand AD coverage for blind and low-vision audiences, complementing rather than replacing human describers. Automatically generated ADs can be contextually plausible but can be factually wrong, which is particularly consequential for users who depend on them as their primary access to visual content; deployment should disclose machine generation and support human review. Both the training data and the underlying VLM carry demographic biases that may surface in generated descriptions, and auditing for descriptive fairness is an important direction we leave to future work. LongLSMDC annotations follow LSMDC's research-only terms; the underlying video content is not redistributed. Released annotations, clips and character banks inherit LSMDC's restricted-access research-only agreement and are distributed through the same request process; code, checkpoints and evaluation scripts are released without restriction. Character banks contain names of fictional characters only and no personal data about real individuals beyond what is already public in the source cast lists.

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In this supplementary material, we provide: (A) dataset, character-bank, and character-name agreement details (Sec. A); (B) fine-grained IoU evaluation, coverage metrics, the CA3D comparison, and the UniAD adaptation protocol (Sec. B); (C) human evaluation (Sec. C); (D) negative-caption examples (Sec. D); (E) Action-Score reproducibility and Gemini prompts (Sec. E); (F) Stage 2 fine-tuning prompts (Sec. F); (G) additional qualitative results and systematic error categories (Sec. G); (H) artifact access and licenses (Sec. H); and (I) model size and budget (Sec. I).

A Additional LongLSMDC and Character-bank details

The original LSMDC (Rohrbach et al., 2017) consists of full-length movies (90–180 minutes) with sparsely distributed AD annotations. Processing full-length movies end-to-end is computationally prohibitive for both training and evaluation: a single 90-minute film contains thousands of frames and hundreds of AD slots, making batched temporal localization infeasible at movie scale. We therefore segment each movie into variable-length clips (5–8 minutes), each containing multiple AD annotations, enabling scalable training and dense multi-segment evaluation.

Release plan. We release: (i) all LongLSMDC annotations—paired visual and spoken windows, clip boundaries, and the train/validation splits; (ii) the preprocessed movie clips and extracted features, under the same terms as LSMDC and in coordination with the original authors; (iii) the per-clip character banks; (iv) the full training and inference code for both stages; (v) the trained Stage 1 localizer and Stage 2 LoRA adapters; and (vi) all evaluation scripts, including the matching protocol and the Action-Score pipeline. Items (i)–(iii) inherit LSMDC's restricted-access, research-only agreement and require the standard LSMDC request; items (iv)–(vi) are unrestricted. Underlying video is not redistributed. Full details and license terms are in Appendix H.

Random clip segmentation. For each movie of duration D, we divide it into non-overlapping clips with randomized durations $\delta \sim \mathcal{U}(300\text{s}, 480\text{s})$. Starting from $t = 0$, we create clip $[t, \min(t + \delta, D))$, then advance t to the clip end, continuing until the entire movie is segmented. Random lengths prevent duration-specific biases and create

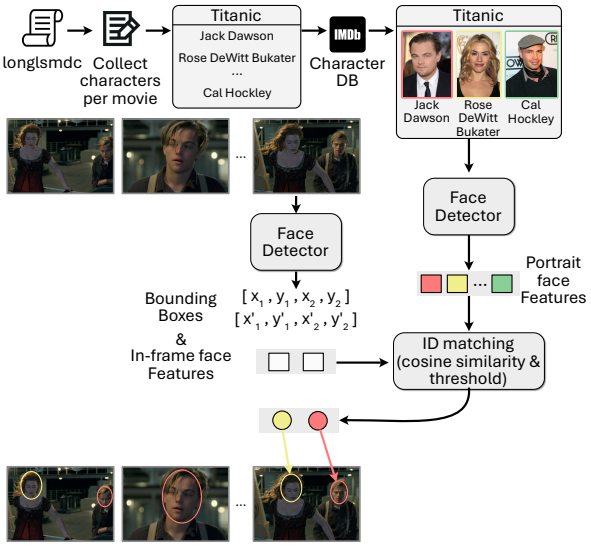


Figure 7: **LongLSMDC character bank construction pipeline.** For each movie, we collect a cast list of character names from an external database. A face detector is applied to movie frames to extract bounding boxes and in-frame face features, which are compared against portrait-level face features for each character via cosine similarity with a fixed threshold, assigning detections to named characters. The resulting per-movie character bank maps frame-level detections to character identities (e.g., Jack Dawson, Rose DeWitt Bukater, and Cal Hockley in *Titanic*), enabling CharBank-conditioned AD generation at inference time.

variable clip complexity (1–147 annotations per clip). The few clips shorter than 300 s (<0.2%) are tail segments produced when a movie ends before the sampled duration expires.

Annotation assignment via midpoint rule. Each AD annotation $[s, e]$ is assigned to clip $c = [t_s, t_e)$ if its temporal midpoint $m = (s + e)/2$ satisfies $t_s \leq m < t_e$. This ensures each annotation belongs to exactly one clip and is temporally centered for maximum context. We compute clip-relative coordinates $s' = s - t_s$ and $e' = e - t_s$ for standardized temporal ranges. Annotations on clip boundaries (<1%) are excluded.

This transforms ∼125 movies into 2,100+ clips (1,995 training, 170 validation) with over 78,000 training annotations across all annotation categories. Individual clips contain between 1 and 147 AD annotations (mean: 29–40 ADs per clip depending on annotation category), providing diverse density conditions for localization evaluation. Dataset density statistics comparing LongLSMDC and CMD-AD are shown in Figure 9: CMD-AD contains short clips (2–3 min)



Figure 8: **Vocabulary comparison between LongLSMDC (Top) and CMD-AD (Bottom) datasets.** LongLSMDC exhibits richer action verb diversity (e.g., "walks", "turns", "looks", "stands") and more varied character descriptors, reflecting its longer clips and broader movie coverage across diverse genres and cinematic styles.

with 1–5 ADs per clip, while LongLSMDC contains longer clips (up to 8 min) with 29–40 ADs per clip, providing substantially more AD events per movie and better supporting evaluation of long-form, multi-segment AD generation.

Character bank construction. Following AutoAD-Zero (Xie et al., 2024), we construct a per-clip character bank for LongLSMDC to support CharBank-conditioned generation experiments (Figure 7). For each movie, we collect a cast list of character names from IMDb. A face detector is applied to movie frames to extract bounding boxes and in-frame face features, which are then compared against portrait-level face features for each character via cosine similarity with a fixed threshold, assigning frame-level detections to named characters. The resulting per-movie character bank maps detected faces to character identities (e.g., Jack Dawson, Rose DeWitt Bukater, and Cal Hockley in *Titanic*), enabling CharBank-conditioned AD generation at inference time. Unlike AutoAD-Zero, which constructs character banks over ground-truth segments, our banks are built over full movie frames and are therefore compatible with predicted windows. Character metadata is provided per clip for both training and validation splits and will be released alongside the dataset.

Character-bank agreement analysis. To measure how often the automatic pipeline finds the character an AD actually talks about, we check, for each character named in a GT AD, whether the bank identifies that character within the AD's window. Of 4,518 character mentions across validation ADs, 748 refer to characters with no portrait in the bank, so no match is possible for them by construction. Among the remaining 3,770 mentions, the bank identifies the character within the window in 2,655 cases (70.4%). We manually inspected a sample of the non-identified cases and found the most frequent cause is not a matching failure at all: the narration references a character whose face is not visible in the window (off-screen, turned away, or shot from behind). No face-matching pipeline, however robust, resolves such cases. For the cases where matching is possible but visually difficult—extreme profiles, occlusion, low lighting—the CharBank comparison in Sec. 5.2 (+3.0 CIDEr with CharBank for Qwen3-VL under GT windows, $34.3 \rightarrow 37.3$, with consistent gains in Action Score and R@1/5) shows that residual matching errors do not propagate harmfully into generation.

Vocabulary analysis. Figure 8 presents word cloud visualizations comparing LongLSMDC and CMD-AD vocabularies. LongLSMDC exhibits richer action verb diversity (e.g., "walks", "turns", "looks", "stands") and more varied character descriptors, reflecting its longer clips and broader movie coverage across diverse genres and cinematic styles. This vocabulary richness supports more comprehensive evaluation of generation models' ability to produce varied, natural language descriptions across different narrative contexts.

Metrics. We follow standard Temporal Action Localization (TAL) evaluation protocols (Tang et al., 2023; Zhang et al., 2022), reporting mean Average Precision (mAP) at temporal IoU thresholds $tIoU \in \{0.2, 0.3, 0.4, 0.5, 0.6\}$. These thresholds are widely used to evaluate TAL methods and assess localization quality at varying precision levels. The temporal IoU (tIoU) is defined as the intersection over union between two temporal windows, equivalent to the 1D Jaccard index. LongLSMDC supplies two ground-truth

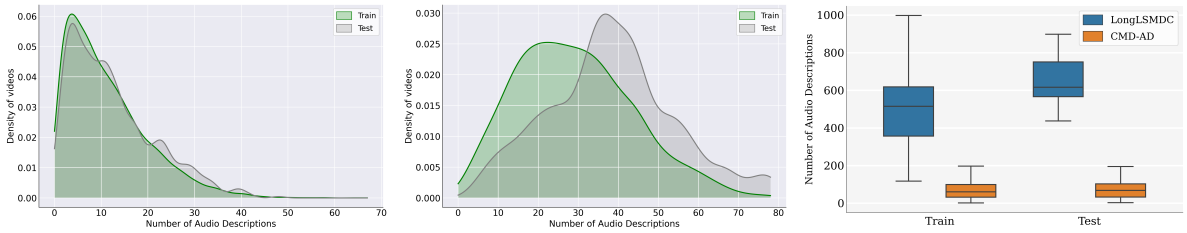


Figure 9: Dataset statistics for CMD-AD and LongLSMDC. (a) CMD-AD contains short clips (2–3 min) with 1–5 audio descriptions per clip. (b) LongLSMDC contains longer clips (up to 8 min) with a much higher density of AD segments (29–40 per clip). (c) Boxplots of the full movie-level distributions show that LongLSMDC provides substantially more AD events per movie than CMD-AD, supporting evaluation of long-form, multi-segment AD generation.

views per opportunity, we report mAP for each head against each view, giving a 2×2 matrix that reveals whether the visual and spoken heads predict the same window or genuinely distinct targets

A predicted segment (t_s^{pred}, t_e^{pred}) matches ground-truth (t_s^{gt}, t_e^{gt}) if their temporal IoU exceeds the threshold:

$$\text{IoU} = \frac{|[t_s^{pred}, t_e^{pred}] \cap [t_s^{gt}, t_e^{gt}]|}{|[t_s^{pred}, t_e^{pred}] \cup [t_s^{gt}, t_e^{gt}]|} \geq \text{threshold} \tag{10}$$

For each ground-truth segment, we follow the greedy matching approach: we find the highest-IoU predicted segment that exceeds the threshold. Matched predictions are considered true positives (TP), unmatched predictions are false positives (FP), and unmatched ground-truth segments are false negatives (FN). Average Precision (AP) is computed per video using the precision-recall curve, and mAP is the mean across all test videos. Given a tIoU threshold, mAP computes the mean of average precision across all videos.

Coverage metrics. Alongside mAP we report the precision, recall and F1 of the predicted spoken narration windows against GT spoken windows under the same greedy one-to-one tIoU matching used for the text metrics. The unmatched-prediction rate is $1 - P$ (predictions with no GT match, i.e., potential false insertions) and the missed-slot rate is $1 - R$ (GT narration slots with no matched prediction). We report both the matching threshold used in the main paper (0.3) and a stricter threshold (0.5); see Table 5.

Coverage interpretation. At tIoU 0.3, recall is 94.7%, leaving 5.3% of GT narration slots unmatched. At the stricter tIoU 0.5 threshold, recall falls to 73.0%, reflecting increased sensitivity to boundary precision. The unmatched-prediction rate can overstate false insertions because Soft-NMS retains overlapping candidates and because LongLSMDC annotates the slots selected by a professional describer rather than every potentially usable dialogue gap. The operating point can therefore be adjusted when insertion-rate control is required.

Table 6: **Multi-seed localization stability on LongLSMDC.** Mean ± standard deviation over three seeds; mAP is averaged over tIoU thresholds 0.2–0.6.

Method	Visual mAP	Spoken mAP	Avg. mAP
Cue2Narrate	42.2±0.1	42.0±0.2	42.1±0.1

Table 7: **Multi-seed generation stability on LongLSMDC.** Qwen3-VL-8B CIDEr, reported as mean ± standard deviation over three seeds.

Evaluation	SFT	$+\mathcal{L}_{drl}$ (Cue2Narrate)
GT→S2	36.0±0.3	37.3±0.2
S1→S2	23.8±0.2	25.2±0.1

For *generation* we use four complementary metrics: **CIDEr** (Vedantam et al., 2015) for n-gram consensus with reference captions, **Recall@**k ($k = 1, 5$) for embedding-space similarity via retrieval, **Action Score** (Xie et al., 2025) for verb correctness (the most informative AD attribute for the listener), and **LLM-Eval** (LLaMA3-8B (Grattafiori et al., 2024) as judge, 1–5 score, mean reported).

B Fine-Grained IoU Analysis for LongLSMDC

Figs. 10 and 11 break down mAP by tIoU threshold $\{0.2, 0.3, 0.4, 0.5, 0.6\}$. Three patterns emerge. First, Gemini 2.5 Pro (Comanici et al., 2025) is competitive at loose tIoU on

LongLSMDC ($\sim$ 12 mAP at 0.2) but collapses sharply as the threshold tightens, falling below 3 mAP at tIoU 0.5 and effectively zero at 0.6. This indicates that prompted timestamp prediction from a general-purpose VLM can identify approximately when an event occurs but cannot place precise temporal boundaries — exactly the property AD localization requires. Second, on LongLSMDC, Cue2Narrate's margin over the unimodal baselines is modest at tIoU 0.2 but widens at stricter thresholds: at tIoU 0.5, Cue2Narrate reaches 30.4/28.6 vs. video-only's 22.8/17.2 and audio-only's 21.8/25.3. This is consistent with the modality roles observed elsewhere in our analysis: audio carries the dominant cue for spoken-slot timing, while video provides the scene-change and action cues needed to disambiguate which silence corresponds to a narratable event (Table 2). Third, the same pattern holds zero-shot on CMD-AD and MAD-Eval, where Cue2Narrate keeps its lead across all thresholds. On CMD-AD, target-dataset fine-tuning has little effect at loose tIoU ($\sim$ 56 vs. $\sim$ 57 at 0.2) but produces a large gain at strict tIoU (FT 42.7 vs. ZS 12.8 at 0.4), showing that fine-tuning primarily improves boundary precision rather than event recall. The dual-head structure transfers asymmetrically by design: the spoken narration window head supervises CMD-AD-style spoken slots, the visual cue window head supervises MAD-Eval-style visual clips, and each head transfers to the benchmark whose window type it was trained against.

B.1 Details of the CA3D F1 Comparison on MAD-Eval

The 90.0 F1 reported for Cue2Narrate on MAD-Eval in Sec. 5.1 follows the top-1 protocol used by CA3D (Lee et al., 2025), which we reproduce here for completeness. MAD-Eval provides exactly one GT visual cue window per clip. For each clip, we retain only the single highest-scoring predicted segment and compute its temporal IoU against the GT window. At a given tIoU threshold θ, a clip contributes as follows: no prediction counts as a false negative (FN $+= 1$); a top-1 prediction with tIoU $\geq \theta$ counts as a true positive (TP $+= 1$); a top-1 prediction below the threshold contributes to both false positives and false negatives (FP $+= 1$, FN $+= 1$), since the prediction is an incorrect detection and the ground-truth window is simultaneously missed. Precision, recall, and F1 are aggregated across all clips as

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \quad R = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \quad F_1 = \frac{2PR}{P+R}. \tag{11}$$

Under this accounting, every missed clip contributes symmetrically to FP and FN, so precision and recall are numerically equal and F_1 coincides with both. Note also that the top-1 restriction is conservative for Cue2Narrate, since additional correct predictions the model produces on multi-event clips are discarded before scoring; the 90.0 vs. 65.3 gap therefore reflects the model's boundary precision on its single strongest detection, not any advantage from producing more candidates.

B.2 UniAD on LongLSMDC

We evaluate UniAD in the Table 3 setting (GT spoken windows) using the released checkpoint pre-trained on MAD, the most closely related available training source to our LSMDC-derived benchmark. Inputs are adapted by converting annotations to clip-relative coordinates and extracting CLIP ViT-L/14 features to match UniAD's expected feature space. UniAD runs without character features, so we compare against our no-CharBank setting. We did not fine-tune UniAD on LongLSMDC; doing so would require train-split conversion to MAD-train format, training-split feature extraction and character-pathway configuration, and we expect it to close part of the reported gap.

C Human Evaluation

To complement automatic metrics, we run a pairwise preference study comparing Cue2Narrate against Shot-by-Shot (Xie et al., 2025) on 100 randomly sampled LongLSMDC clips. For each clip, 20 evaluators are shown the video alongside two anonymised AD tracks (ours and the baseline, order randomised) and asked to choose the preferred track on three axes: (1) **Coverage of key moments** whether the ADs cover the salient on-screen actions and interactions; (2) **Temporal order & coherence** whether the ADs appear at appropriate moments and follow the story flow; and (3) **Narrative quality & clarity** whether the ADs read as coherent, listener-friendly narration. Fig. 12 reports the percentage of pairwise judgments preferring each system. Cue2Narrate is preferred on every axis, with the largest gap on nar-

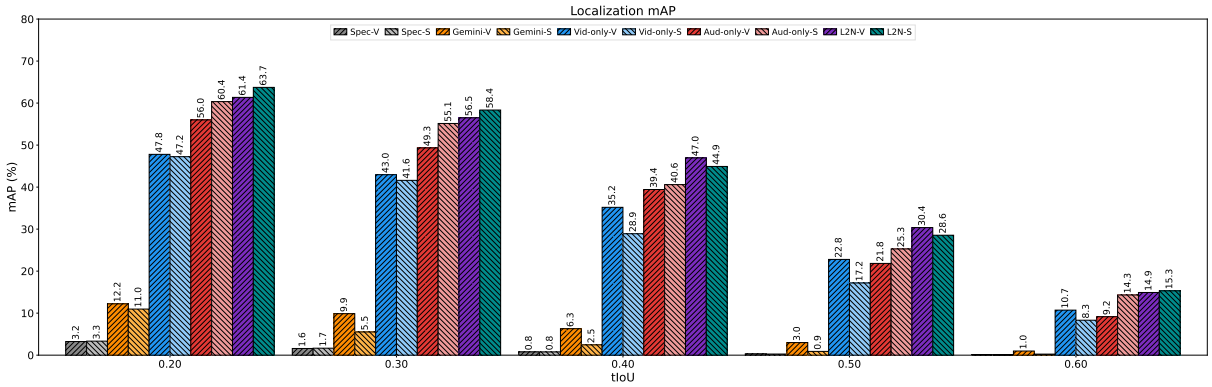


Figure 10: **Per-tIoU localization on LongLSMDC.** Grouped bar plot of mAP at tIoU thresholds $\{0.2, 0.3, 0.4, 0.5, 0.6\}$ (x-axis), with bar height encoding mAP (%). For each threshold, ten bars compare five methods evaluated against both the visual cue window head (**-V**, hatched) and the spoken narration window head (**-S**, solid): Spectrogram (Spec), Gemini 2.5 Pro, video-only ActionFormer (Vid-only), audio-only ActionFormer (Aud-only), and Cue2Narrate (C2N). Gemini is competitive at loose tIoU but collapses at stricter thresholds; Cue2Narrate leads across all thresholds, with the margin over unimodal baselines widening as tIoU tightens.

rative quality, consistent with the AD-style framing the Description Ranking Loss is designed to encourage. Our evaluators are sighted by design: the three axes require comparing the AD track against the visual ground truth, which is only possible with visual access. This makes the study complementary to, not a substitute for, a listening-experience study with blind and low-vision users; see Sec. 7.

D Examples of negative captions.

To construct the negatives used by DRL (Sec. 4.3), we prompt the frozen Qwen3-VL backbone with a simple prompt (shown in the box below) on each training clip and cache the resulting caption offline. The same prompt format is used across all clips, so the negatives reflect the backbone's default captioning prior long, descriptive, scene-summarising rather than any AD-style framing. Two representative pairs of (caption negative, ground-truth AD) are shown in Fig. 13; the detailed-vs-concise gap visible in both examples is what DRL is trained to exploit.

System prompt. You are an intelligent chatbot designed for generating audio descriptions for movies. Audio Description (AD) provides concise spoken narration of important visible events, actions, characters, and scene changes in a movie. The prompt includes several short examples of valid AD sentences.

User prompt. `<video>` Watch the following movie clip and generate an Audio Description for it.

Constraints. The generated AD should be exactly one sentence, use present tense, describe only visible information, avoid hallucination, focus on important people, actions, and interactions, and avoid naming characters unless their identity is visually explicit or provided.

E Gemini prompt details for Action-Score metric

Reproducibility of the Action Score. Action Score is an existing metric introduced by Shot-by-Shot; computing it on LongLSMDC requires no access to closed-source software. The closed-source model is used at exactly one point: offline preprocessing of the GT annotations, replacing character names with pronouns and splitting each AD into single-action subsentences (prompts below). We release the resulting GT action annotations alongside the dataset—the same practice as Shot-by-Shot, which ships its precomputed GT action files with its code. Given these files,

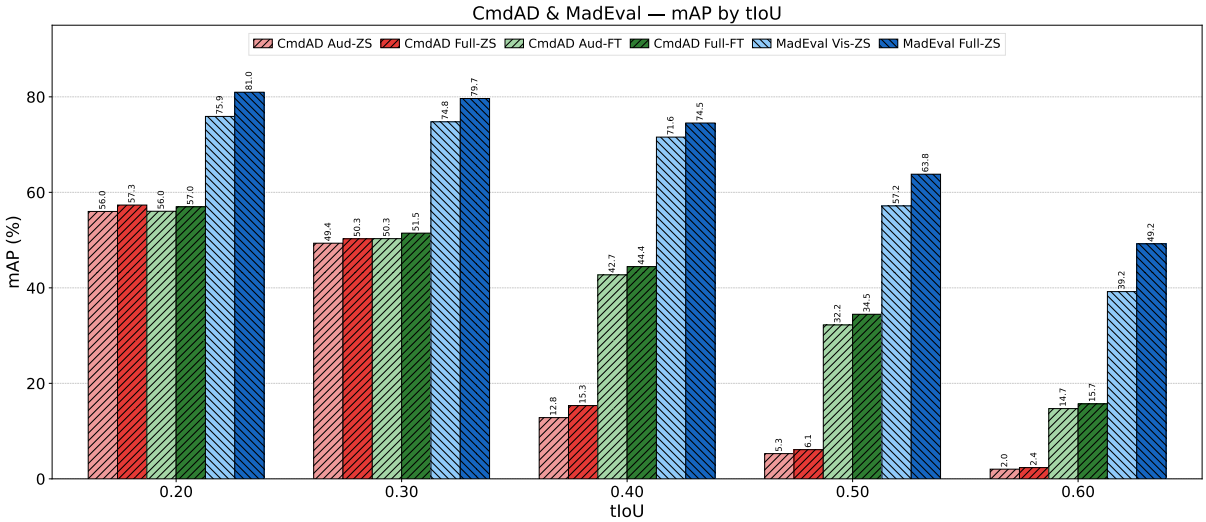


Figure 11: **Per-tIoU transfer to CMD-AD and MAD-Eval.** Grouped bar plot of mAP at tIoU thresholds $\{0.2, 0.3, 0.4, 0.5, 0.6\}$ on the two transfer benchmarks. CMD-AD reports spoken narration window mAP with audio-only (**Aud**, red) and the full Cue2Narrate (**Full**, dark red) under both zero-shot (**ZS**, hatched) and fine-tuned (**FT**, solid) settings. MAD-Eval reports visual cue window-window mAP for video-only (**Vis-ZS**, light blue, hatched) and the full Cue2Narrate (**Full-ZS**, dark blue, solid), evaluated zero-shot. On CMD-AD, ZS and FT are nearly identical at loose tIoU but FT pulls sharply ahead at stricter thresholds, showing that target-dataset fine-tuning primarily sharpens boundary precision. Cue2Narrate's lead persists across all thresholds and both transfer benchmarks.

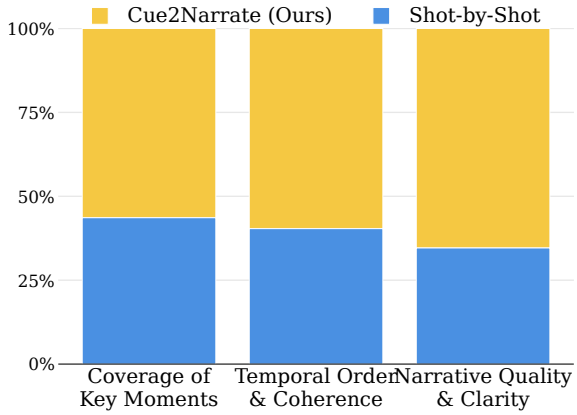


Figure 12: **Human evaluation on LongLSMDC.** Pairwise preference between Cue2Narrate (yellow) and Shot-by-Shot (Xie et al., 2025) (blue) across three dimensions. Evaluators preferred Cue2Narrate on all three axes, with the largest margin on Narrative Quality & Clarity (∼65% vs. 35%).

evaluating any prediction is fully reproducible: the predicted AD is parsed with rule-based dependency parsing (no model calls) and compared to the GT actions with an open-source sentence-embedding model (GTE) plus verb matching. No paid or proprietary model is needed to evaluate on LongLSMDC.

Example 1 `train_3001_21_JUMP_STREET_clip0_176.321_178.664.mp4`

Caption negative: *The video begins with a close-up shot of a young man with short, light-colored hair, wearing a white shirt. He is sitting on a bench, looking off to the side with a serious expression. The background shows a brick wall and a green bench, suggesting an outdoor setting, possibly a school or public area. The scene then transitions to a wide shot of the exterior of a building, which appears to be a school or institutional facility, with a glass door marked "School of Business." Two individuals are walking away from the entrance.*

Ground-truth AD: Sitting on a nearby bench, the nerd looks over.

Example 2 `train_3001_21_JUMP_STREET_clip0_179.061_182.119.mp4`

Caption negative: *The video opens with a wide shot of the exterior of a school building, where two students are standing outside. One student, wearing a purple and white jacket, is leaning against the wall with their head down, appearing to be in distress or deep thought. The camera then zooms in on the student, who continues to lean against the wall, occasionally wiping their face with their hand, suggesting they might be crying.*

Ground-truth AD: The jock slumps down on a bench on the other side of two columns.

Figure 13: **Caption negatives used by the Description Ranking Loss.** For each training clip, a frozen LLM Qwen3-VL produces a long, descriptive caption (italicized) that DRL pushes the model away from in favor of the concise ground-truth AD.

System Prompt

<role>

You are an intelligent chatbot designed for removing character information of a sentence.

</role>
<instruction>

Here 's how you can accomplish the task : You should replace all character information including names , roles , and jobs into pronouns (e.g., he , she , they , her , him , them). Note , objects, locations , and animals are not counted as character information and should be kept as - is .

</instruction>
<output_format>

You should output the result in JSON format WITHOUT providing ANY more sentences at the beginning or at the end .

</output_format>

User Prompt

Please read the sentence below that describes a video clip:
Input sentence: <text_gt>
Replace all character information including names , roles , and jobs into pronouns (e.g., he , she , they , her , him , them). Note , objects, locations , and animals are not counted as character information and should be kept as - is .
Here are some examples:

- Example1: Input sentence : Spicoli watches Mr . Hand pass out the schedule. Ouput : He watches him pass out the schedule.
- Example 2: Input sentence : Waiting for a reply , the inspector has a look of smug satisfaction as he combs his neat moustache. Output: Waiting for a reply , he has a look of smug satisfaction as he combs his neat moustache.
- Example 3: Input sentence : Emmerich 's eyebrows twitch as he watches her. Output : His eyebrows twitch as he watches her.
- Example 4: Input sentence : Inside is a second pair of doors. Output : Inside is a second pair of doors.
- Example 5: Input sentence : The blonde saunters over to him in her striped pantsuit and leans in for a kiss. Output : She saunters over to him in her striped pantsuit and leans in for a kiss.

System Prompt

<role>

You are an intelligent chatbot designed for decompose the sentence into subsentences .

</role>
<instruction>

Here 's how you can accomplish the task : You should split (rewrite if needed) the sentence into subsentences , each containing only one action phrase (i.e., verb phrase).

</instruction>
<output_format>

You should output your answer in JSON format WITHOUT providing ANY more sentences at the beginning or at the end .

</output_format>

Then the sentence with the removed character is used as input, and the ground truth actions are extracted:

User Prompt

Please read the sentence below that describes a video clip:
Input sentence : <text_wo_char>
Split and rewrite the sentence into subsentences , each containing only one action (i.e.,verb phrase) and preserving all other information (e.g., locations , time , affections , etc .) . Do not output repeating actions.
Examples :

- Example1: Input sentence : He watches him pass out the schedule . Subsentences : [He watches him ., He passes out the schedule .]"
- Example2: Input sentence: Waiting for a reply , he has a look of smug satisfaction as he combs his neat moustache . Subsentences : [He waits for a reply ., He has a look of smug satisfaction ., He combs his neat moustache .]
- Example3: Input sentence : He swings in front of Kingpin , then bounces off a building and kicks the criminal into the air . Subsentences : [He swings in front of him ., He bounces off a building ., He kicks him into the air .]
- Example4: Input sentence : His eyebrows twitch as he watches her . Subsentences : [His eyebrows twitch ., He watches her .]
- Example5: Input sentence : Inside is a second pair of doors . Subsentences : [Inside is a second pair of doors .]

F Cue2Narrate (Stage 2) Fine-tuning Prompts

For fine-tuning the Cue2Narrate (Stage 2) module, we employ structured prompts to generate audio descriptions conditioned on localized temporal boundaries. Prompts `System Prompt` and `User Prompt` show the complete prompt design used during fine-tuning.

System Prompt

You are an intelligent chatbot designed for generating audio descriptions for movies. Audio description (AD), also referred to as a video description, described video, or visual description, is a form of narration used to provide information surrounding key visual elements in a media work for the benefit of blind and visually impaired consumers. Audio descriptions are typically placed within a gap in dialogue or important sound.
Here are some examples of audio descriptions that were generated for a different movie clip:

- Example 1: Sitting on a nearby bench, the nerd looks over.
- Example 2: Now students spar in a hand-to-hand combat class.
- Example 3: He slams him down on the mat and raises his fist.
- Example 4: She stares distantly, then faces Andy.
- Example 5: A young girl cycles through the city at night.

The `System Prompt` establishes the model's role as an audio description generator and defines the output format requirements, ensuring consistent structure across all generated descriptions. The `User Prompt` provides the necessary context including video metadata, temporal boundaries from the localization stage, and ground-truth captions to guide the generation process. This two-stage prompting strategy enables the model to learn the mapping from localized temporal segments to natural language audio descriptions.

User Prompt

<video>

Watch the following movie clip and generate an Audio Description for it. The Audio Description should only consist of one sentence. Make sure you do not hallucinate information and only describe what you can see in the movie clip.
Requirements:

- Describe only the most important visible information in the clip.
- Focus on the visible people or subjects, their actions, and important interactions.
- Use present tense.
- Do not name characters unless their identity is explicitly provided in the input or visually confirmed. Instead use neutral words such as "A man", "A woman", "He", "She" etc. Assistant target the GT AD sentence, e.g.: Sitting on a nearby bench, the nerd looks over.

G Additional Qualitative Results

G.1 Systematic error categories

We organise generation errors into four categories: wrong-subject attribution, an under-specified verb despite identifying the correct event, boundary drift that describes adjacent content, hallucinated visual detail, and correct content expressed with wording that differs from the reference. The main-paper example "Ace stands and watches Chris walk away" versus "Gordy pulls the hammer" illustrates wrong-subject attribution when several plausible visual subjects share the frame. Figures 14–17 provide representative boundary-drift, under-specification, hallucination, and alternative-phrasing cases. This variability is consistent with the inter-annotator variability of AD reported in prior work.

Figures 14 and 15 illustrate our model's localization and generation performance on CMD-AD test clips, while Figures 16 and 17 show results on LongLSMDC. Each example shows: (top) sampled video frames from the continuous clip, (middle) mel-spectrogram visualization where bright regions indicate audio activity and dark regions indicate silence periods suitable for AD insertion, and (Bottom) temporal alignment where black bars represent ground-truth AD segments and blue bars show our predicted segments.

CMD-AD Results: Learning Selective Silence Awareness. Figure 15 demonstrates that our proposed localization approach learns not every silence period requires an AD. In the top example, the mel-spectrogram shows multiple silence gaps (dark regions) throughout the 25-second clip, yet our model correctly predicts only 4 AD segments that align with ground truth. This selectivity reveals that the model distinguishes between acoustically silent moments that are narratively important versus those that are not. The audio-visual fusion enables this discrimination while audio detects silence periods, the visual stream determines whether the corresponding visual content is salient enough to describe. For instance, brief pauses during continuous action may be acoustically silent but visually redundant, requiring no narration. The predicted segments (blue bars) consistently align with both mel-spectrogram silence regions (dark bands) and ground-truth annotations (black bars), confirming that our approach successfully learns the joint constraint of acoustic availability and visual salience.

Table 8: **Artifact release plan.**

Artifact	Access	Terms
LongLSMDC annotations (paired windows, splits)	Request	LSMDC research-only
Preprocessed clips + extracted features	Request	LSMDC research-only
Per-clip character banks	Request	LSMDC research-only
GT action annotations (Action Score)	Request	LSMDC research-only
Stage 1 + Stage 2 training/inference code	Public	Open license
Trained checkpoints (localizer, LoRA adapters)	Public	Open license
Evaluation scripts (matching, all metrics)	Public	Open license

LongLSMDC Results: Dense, Continuous Annotations. Figures 16 and 17 showcase performance on densely annotated sequences where ADs occur in rapid succession, reflecting real-world untrimmed movie scenarios. Figure 17 is particularly challenging: 4-5 adjacent ground-truth AD segments occur within a 20-second window (timestamps 210-260s), with minimal gaps between them. This represents the high-density scenario characteristic of LongLSMDC, where ADs must be inserted continuously during extended dialogue-free sequences. Our model maintains temporal precision despite this tight spacing, with predicted segments (blue bars) closely tracking ground truth across the rapid sequence. The mel-spectrogram reveals sustained low-activity regions (dark purple bands) spanning multiple seconds, and our proposed localization approach successfully partitions these extended silence periods into multiple discrete AD segments based on visual changes. This demonstrates robust generalization to the challenging case of continuous narrative coverage where the model must identify not just isolated silence gaps, but also how to temporally segment extended silent periods into appropriate segments. Some prediction errors occur due to the extreme difficulty of these scenarios, highlighting the challenges of dense, real-world movie description tasks.

H Artifact License

The datasets and models employed in this work are subject to the following licenses. The Qwen2.5 and Qwen3 model families are released under the Apache 2.0 License, which permits free use, modification, and distribution for both research and commercial purposes. The Condensed Movies Dataset (CMD) is distributed under the Creative Commons Attribution 4.0 International License (CC BY 4.0), which allows unrestricted use provided appropriate credit is given. The Large Scale Movie Description Challenge (LSMDC) dataset is made available under a restricted-access agreement through the Max Planck Institute for Infor-

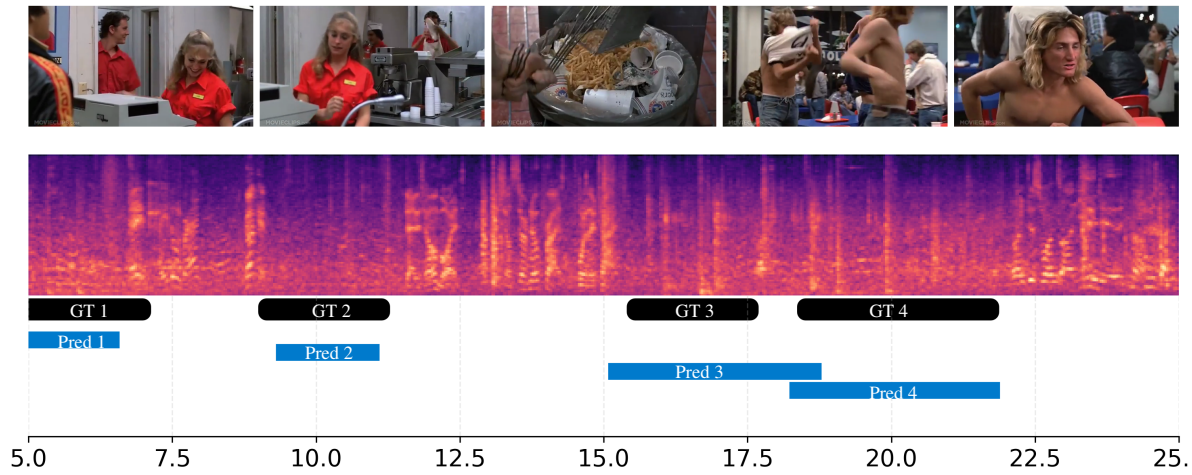


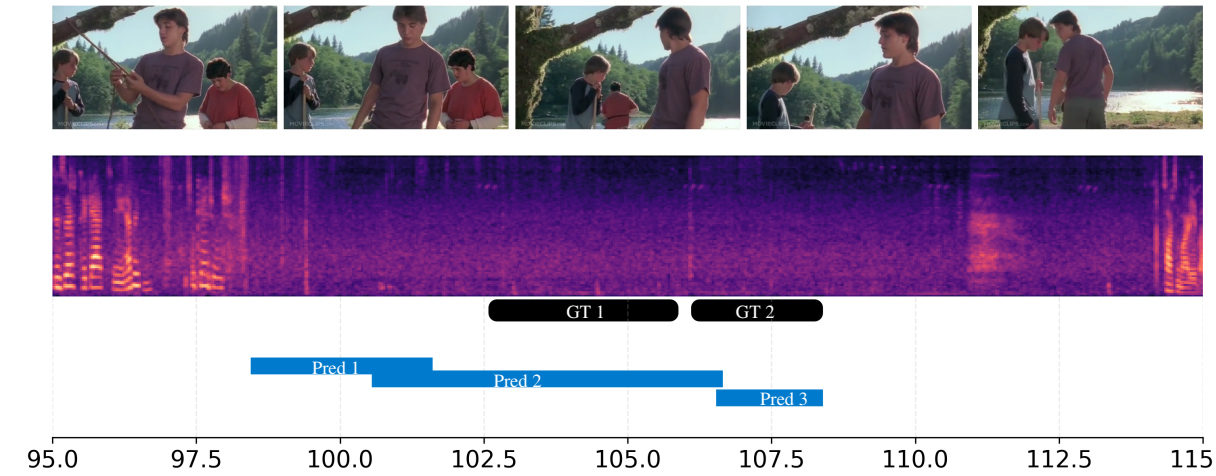
Figure 14: **CMD-AD qualitative examples on curated short clips.** Our model demonstrates effective localization and generation on moderately dense sequences. (Top) Sampled video frames, (Middle) mel-spectrogram showing audio activity (bright regions) and silence periods (dark regions), (Bottom) temporal alignment with ground-truth segments (black bars) and predicted segments (blue bars). The model successfully identifies appropriate silence windows and generates contextually relevant descriptions. GT: Ground Truth, Pred: Prediction.

matics (MPII), permitting use exclusively for non-commercial academic research. The MAD-Eval (Movie Audio Descriptions) dataset is governed by a Non-Disclosure Agreement (NDA) that limits its use to academic research purposes only. All resources are therefore used in full compliance with their respective licensing terms within the scope of this academic work.

I Model size and budget

Stage 1. The audio-visual dual-head localizer has 6.64M parameters in total, shared across all three variants (*video-only*, *audio-only*, *full*); mode flags disable compute paths but do not change parameter count.

Stage 2. We use Qwen2.5-VL-7B or Qwen3-VL-8B as the backbone, adapted with LoRA (r=16, α=32, dropout 0.05) on the language model and multimodal projection; the vision encoder is frozen.

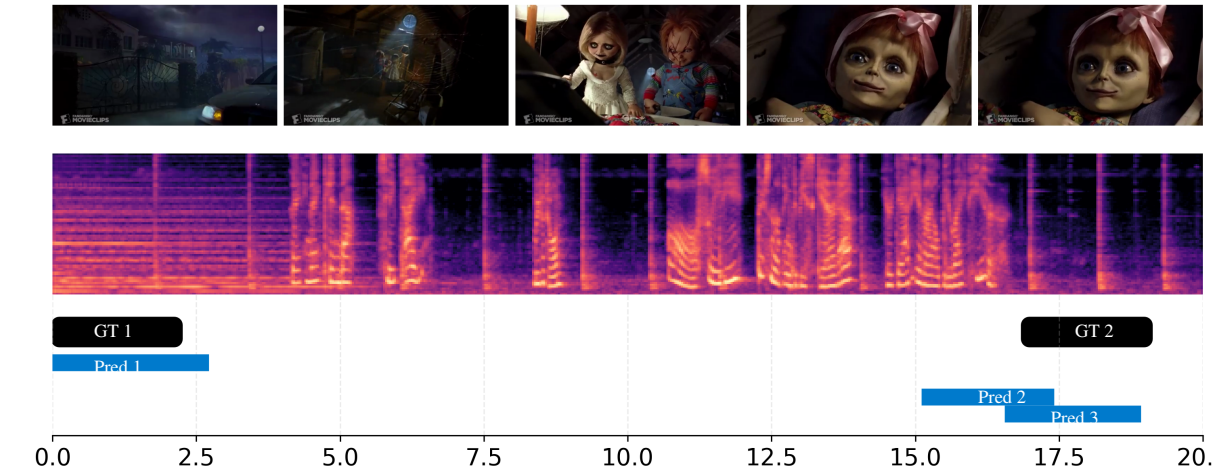


GT 1: George packs up his Swiss Army knife and plods away.

Pred 2: George folds up his knife and slowly walks away.

GT 2: Sam and Rocky watch him walking away.

Pred 3: He stands on one leg, then the other.



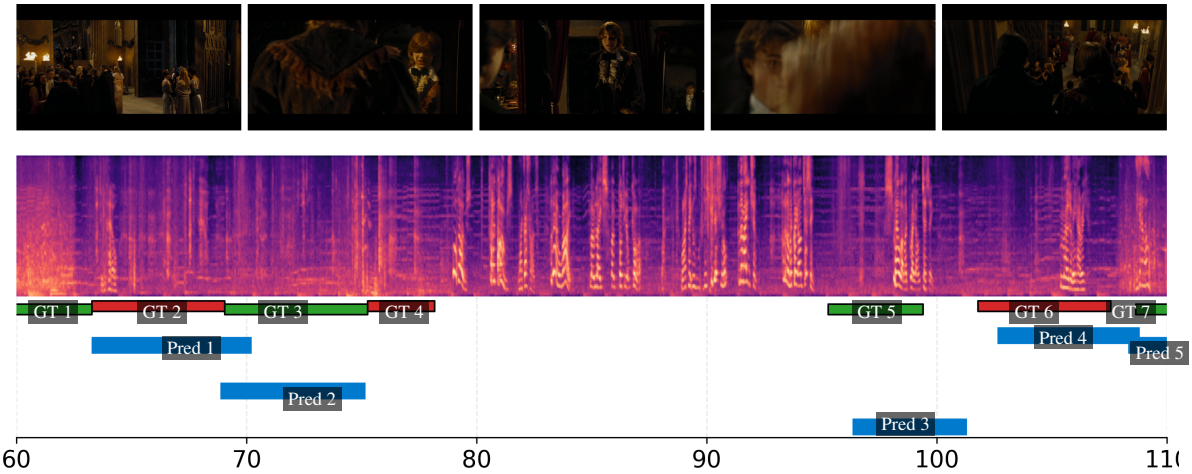
GT 1: Tiffany reaches for the lamp.

Pred 1: The light shines through the window above the attic door.

GT 2: Glen, Glenda stares at Chucky.

Pred 2: Tiffany and Chucky look at each other.

Figure 15: **CMD-AD example demonstrating selective silence awareness.** The model learns that not every acoustic silence requires an audio description. In the top example, multiple silence gaps appear in the mel-spectrogram throughout the 25-second clip, yet the model correctly predicts only segments corresponding to visually salient events that align with ground truth. This demonstrates that audio-visual fusion enables discrimination between acoustically silent moments that are narratively important versus those that are not. The visual stream determines content salience while audio detects insertion opportunities. GT: Ground Truth, Pred: Prediction.



GT 1: The girls wear long gowns, the boys dress robes.

GT 2: In his room, Ron adjusts his tatty robe.

Pred 1: HARRY is in the Great Hall, at the Gryffindor table.

GT 3: Ron stares at himself in a mirror.

Pred 2: The queen turns to her son.

GT 4: Harry comes in wearing a smart black robe and a white bow tie.

GT 5: Harry takes a deep breath and sinks down after it.

Pred 3: They look at each other.

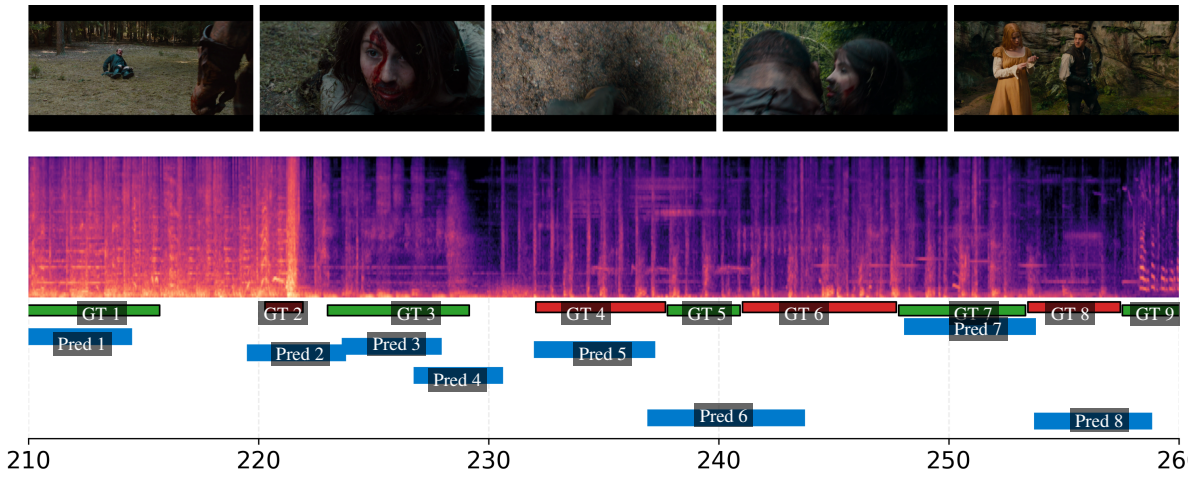
GT 6: Moody searches the shelves.

Pred 4: They are all wearing costumes.

GT 7: Snowflakes whirl outside the boys' bedroom.

Pred 5: They both wear bow ties as it snows.

Figure 16: **LongLSMDC example showing dense annotations in real-world scenarios.** LongLSMDC presents significantly more challenging scenarios with high annotation density and longer, untrimmed clips. This example shows 7 ground-truth segments within a 50-second window, demonstrating rapid succession of events typical of continuous movie sequences. The model maintains temporal precision despite tight spacing, with predicted segments closely tracking ground truth. The mel-spectrogram reveals sustained silence periods that must be partitioned into multiple discrete AD opportunities based on visual content changes, demonstrating the model's ability to handle continuous narrative coverage. GT: Ground Truth, Pred: Prediction.



GT 1: Mina stands with her wand ready after assaulting Muriel with a spell.

Pred 1: A man with long hair and a beard, dressed in leather, and carrying a spear, lifts a woman into his arms.

GT 2: The troll stomps on his head, his brains and blood splattering on the ground.

Pred 2:

GT 3: Mina shatters a tree behind Muriel.

Pred 3: The man stands in the middle of a clearing.

Pred 4: The giant turns and sees Jane.

GT 4: Gretel awakens to the gentle rocking motion of being carried.

Pred 5: The light from the flashlight beam moves about the ground.

GT 6: Gretel turns her head to find the side of the troll's head.

GT 5: Ben watches Gretel fall.

Pred 6: John reaches out to grab the giant's foot.

GT 7: Gretel closes her eyes and leans her head to rest against the giant's shoulder.

Pred 7: A man with long hair and a beard, dressed in leather, and carrying a spear, lifts a woman into his arms.

GT 8: They kiss.

Pred 8: Muriel and her other witch, one with red hair, hover menacingly.

Figure 17: **LongLSMDC example showing dense annotation and real-world annotation scenarios.** This sequence contains 9 ground-truth segments within a 50-second window (4-5 adjacent segments in rapid succession around timestamps 210-260s), representing the most challenging high-density scenarios in real-world untrimmed movies. The model must handle: (1) minimal gaps between consecutive ADs, (2) complex visual scenes with multiple characters and actions, (3) extended dialogue-free sequences requiring continuous narrative coverage. Despite these challenges, the localizer successfully partitions extended silence periods into appropriate description units. Some prediction errors occur due to the extreme difficulty of this scenario, highlighting areas for future improvement. GT: Ground Truth, Pred: Prediction.